

Complementi di Circuiti e sistemi elettronici

Anno accademico 2024/2025

Soluzioni

(3) 1. $\frac{V_{out}(s)}{V_{in}(s)} = \frac{sC(R_2 + R_1) + 1}{sCR_2 + 1}$

(7) 2.

$$V_{gs1,2,3e} = 1.1V$$

$$V_{gs3,4,5,6} = 1.1V$$

$$V_a = 0.7 + 1.1 = 1.8V$$

$$V_b = 5 - 2.2 = 2.8V$$

$$V_{out} = V_{in} = 0.7V$$

$$V_c = 0.7V + 1.1 = 1.8V$$

(2) 3. $G_{loop}(0) = - \frac{R_{eq}}{1/gm_7 + 1/gm_2 + R_1}$

$$R_{eq} = r_{04} + r_{06}(1 + gm_6 r_{04}) = 122 M\Omega$$

$$G_{loop}(0) = -1830$$

(7) 4.

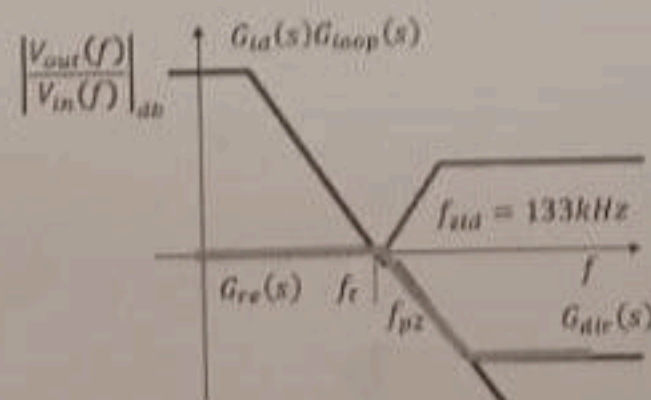
$$G_{loop}(s) = \frac{G_{loop}(0)(1 + s\tau_{z1})}{(1 + s\tau_{p1})(1 + s\tau_{p2})}$$

$$f_{p1} \cong 65 Hz$$

$$f_{p2} \cong 460 kHz$$

$$f_z \cong 800 kHz$$

$$f_t \cong G_{loop}(0)f_{p1} = 119 kHz$$



(7) 5.